STRUCTURAL MODELING OF HETEROGENEOUS DATA WITH PARTIAL LEAST SQUARES

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ABSTRACT

Alongside structural equation modeling (SEM), the complementary technique of partial least squares (PLS) path modeling helps researchers understand relations among sets of observed variables. Like SEM, PLS began with an assumption of homogeneity – one population and one model – but has developed techniques for modeling data from heterogeneous populations, consistent with a marketing emphasis on segmentation. Heterogeneity can be expressed through interactions and nonlinear terms. Additionally, researchers can use multiple group analysis and latent class methods. This chapter reviews these techniques for modeling heterogeneous data in PLS, and illustrates key developments in finite mixture modeling in PLS using the SmartPLS 2.0 package.

1. INTRODUCTION

Structural modeling techniques such as structural equation modeling (SEM) or partial least squares (PLS) path modeling incline researchers to a high

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level of abstraction, focusing on latent variables or linear composites, rather than on the observations themselves. Yet data concerns, such as the presence of outliers (Bollen & Arminger, 1991), are still important issues for structural modeling, as much as they are for regression and other techniques. Applications of structural equation models are usually based on the assumption that the analyzed data stem from a single population, so that a unique global model represents all the observations well. However, in many real-world applications, this assumption of homogeneity is unrealistic. Heterogeneous perceptions and evaluations of products and services form the basis of the concept of market segmentation. Researchers long ago noted the importance of considering heterogeneity in SEM (e.g., Capecchi, 1973). Including interactions and nonlinear terms in a model is one way to allow for heterogeneity across respondents. These higher order terms mathematically imply that simple slopes – the derivative of the dependent variable with respect to a given predictor – vary by respondent, depending on the respondent's score on another predictor (Cohen, Cohen, West, & Aiken, 2003). Traditionally, however, heterogeneity implies the existence of distinct groups of respondents. These groups may be defined a priori on the basis of, for instance, geographic variables or stated preferences, but heterogeneity is frequently unobservable and its true sources are unknown to the researcher (Wedel & Kamakura, 2000). Based on the seminal work by Blåfield (1980), finite mixture models treat group identity as a latent variable to be discovered from the data. Covariance-based SEM has long included multiple group analysis (Jöreskog, 1970) to handle a priori groups. SEM also includes hierarchical or multilevel models for clustered data (Stapleton, 2006; Wetzels, Odekerken-Schöder, & van Oppen, 2009). Likewise, finite mixture modeling has been applied to SEM (Arminger & Stein, 1997; Jedidi, Jagpal, & DeSarbo, 1997a; Yung, 1997; Dolan & van der Maas, 1998; Arminger, Stein, & Wittenberg, 1999; Lee & Song, 2002). SEM researchers have also fused multilevel modeling and mixture modeling into hybrid multilevel growth mixture modeling procedures (e.g., Muthén, 2002, 2008).

Likewise, the PLS approach to structural modeling has come to include approaches for dealing with heterogeneity (Henseler, Ringle, & Sinkovics, 2009). Originally developed by Wold (1974, 1982) and further extended by Lohmöller (1989), PLS path modeling represents an alternative to covariance-based SEM. PLS avoids most distributional assumptions, explicitly fortifies exploration of alternative models, and works dependably even at relatively low sample sizes (Cassel, Hackl, & Westlund, 1999; Reinartz, Haenlein, & Henseler, 2009) and formative measurement models (Ringle, Götz, Wetzels, & Wilson, 2009). PLS path modeling has lately established itself as a tool for researchers, especially in the management information systems (MIS) and marketing disciplines. The use of PLS path modeling in MIS mainly draws on Davis' (1989) technology acceptance model (e.g., Agarwal & Karahanna, 2000; Gefen & Straub, 1997; Igbaria, Zinatelli, Cragg, & Cavaye, 1997). In marketing, the various customer satisfaction index models, such as the American Customer Satisfaction Index (ACSI; Fornell, Johnson, Anderson, Cha, & Bryant, 1996), represent a key area for applying the PLS path modeling methodology.

Tools for dealing with heterogeneous data within PLS path modeling have grown rapidly, with many new developments appearing only in specialized literature devoted to PLS methods. The aim of this chapter is to provide a coherent review of these developments for a broader audience. Furthermore, we illustrate the general problem of structural modeling of heterogeneous data by taking a closer look at finite mixture PLS (FIMIX-PLS; Hahn, Johnson, Herrmann, & Huber, 2002), the most prominent approach for treating unobserved heterogeneity in a PLS framework (Sarstedt, 2008a). The remainder of this chapter proceeds as follows. First, the chapter provides a brief overview of conventional PLS path modeling for a homogeneous population. Then it looks at three distinct approaches for accounting for heterogeneity: single group models with nonlinear and interaction terms, multiple group models with a priori defined groups, and multiple group models with classes inferred based on model results. The chapter offers a detailed empirical example of the latter approach, an application of the FIMIX-PLS method to customer satisfaction and reputation data. It concludes with some general recommendations.

2. PARTIAL LEAST SQUARES PATH MODELING

Following is a brief review of the essentials of PLS path modeling. In applications, the SmartPLS (Ringle, Wende, & Will, 2005b) and PLS-Graph (Soft Modeling, Inc., 1992–2002) software provide easy access to the multivariate analysis technique by intuitive graphical user interfaces. For a more extensive introduction, see, for example, Chin (1998), Haenlein and Kaplan (2004), Henseler et al. (2009), and Hair, Ringle, and Sarstedt (2011).

As in typical SEM analyses, PLS path modeling associates sets of observed variables with other variables, which may be considered latent variables because they are not openly present in the data set. Unlike SEM, however, PLS uses latent variable proxies which are linear composites of the associated observed variables. These latent variable proxies in PLS can always be expressed as exact linear functions of their indicators.

In general, a PLS path model consists of latent variables and their observed variable indicators. The latent variables may be divided into an $M \times 1$ vector of endogenous constructs η and a $J \times 1$ vector of exogenous constructs ξ, which are linked by the structural equation (Table A.1 provides a description of all of the symbols used in the equations presented in this chapter):

$$\eta = B\eta + \Gamma\xi + \zeta \tag{1}$$

Here, ζ is an $M \times 1$ vector of residuals, normally distributed, mutually uncorrelated with ζ_m $(m = 1, \ldots, J_M)$, the residual for construct η_m $(m = 1, \ldots, M)$, uncorrelated with all predictors in the η_m structural equation. This set of equations comprises the "inner model," while measurement equations, linking constructs with observed variables, comprise the "outer model." Every η_m construct is associated with one or more observed variables y_m, and every ξ_j construct is associated with one or more observed variables x_j. The nature of these relationships depends on the "mode" chosen by the researcher. In Mode A, these relationships are expressed as single regressions (Wold, 1982):

$$x_j = w_j\xi_j + \delta_j, \quad \text{for each } x \text{ in the } j \text{ subvector} \tag{2}$$

$$y_m = w_m\eta_m + \delta_m, \ \text{for each } y \text{ in the } m \text{ subvector} \tag{3}$$

The roles are reversed in the multiple regression models for Mode B:

$$\xi_j = \sum(w_jx_j) + \delta_j, \ \text{for all } x \text{ in the } j \text{ subvector} \tag{4}$$

$$\eta_m = \sum_{m=1}^{M}(w_my_m) + \delta_m, \ \text{for all } y \text{ in the } m \text{ subvector} \tag{5}$$

In both modes, error terms δ are normally distributed with 0 means and are uncorrelated with the predictors in their respective equations. In practice, "reflective measurement" in the PLS literature is synonymous with Mode A, while "formative measurement" is synonymous with Mode B. If different modes are chosen for different latent variables, the literature labels this Mode C. The researcher may choose different modes for different latent variables, but may not mix modes for a given latent variable.

In PLS path modeling, parameter estimation is accomplished through a multistage algorithm. The various stages involve a sequence of regressions in terms of weight vectors, with iteration leading to convergence on a final set of weights (Henseler, 2010). The weight vectors obtained at convergence satisfy fixed point equations; see Dijkstra (1981, 2010) for a general analysis of such equations and ensuing convergence issues. The basic PLS algorithm, as suggested by Lohmöller (1989), includes three stages which Henseler et al. (2009) summarize as follows:

Stage 1: Iterative estimation of latent variable scores; steps #1.1–1.4 are repeated until convergence is obtained.

Step 1.1: Outer approximation of the latent variable scores. Outer proxies of the latent variables are calculated as linear combinations of their respective indicators. These outer proxies are standardized; for example, they have a mean of zero and a standard deviation of one. For the initial iteration, any arbitrary nontrivial linear combination of indicators can serve as an outer proxy of a latent variable. In practice, equal weights are a typical choice. Later iterations use weights from Step 1.4. These proxies are standardized to 0 means and unit variance.

Step 1.2: Estimation of the inner weights. In the next step, the algorithm constructs a new inner model proxy for each latent variable as a weighted sum of the outer model proxies, constructed in the previous step, for those latent variables which are directly connected to the latent variable in question. There are three schemes available for determining these inner weights. Wold (1982) originally proposed the centroid scheme. Later, Lohmöller (1989) developed the factor weighting and path weighting schemes. The centroid scheme simply uses unit weights adjusted for the sign of the correlation between the two proxies. The factor weighting scheme uses the correlations themselves as the weights. The path weighting scheme pays tribute to the arrow orientations in the path model. The weights of those latent variables that explain the focal latent variable are set to the regression coefficients yielded from a regression of the focal latent variable (regressant) on its latent repressor variables. The weights of those latent variables that are explained by the focal latent variable are determined in a similar manner as in the factor weighting scheme. Regardless of the weighting scheme, a weight of zero is assigned to all nonadjacent latent variables. While these schemes seem very different, practice suggests that the choice makes little difference (Henseler et al., 2009).

Step 1.3: Inner approximation of the latent variable scores. Inner proxies of the latent variables are calculated as linear combinations of the outer proxies of their respective adjacent latent variables, using the afore-determined inner weights. Again, the proxies are standardized to 0 means and unit variance.

Step 1.4: Estimation of the outer weights. The outer weights are calculated either as the covariances between the inner proxy of each latent variable and its indicators (in Mode A) or as the regression weights resulting from the ordinary least squares regression of the inner proxy of each latent variable on its indicators (in Mode B).

Stage 2: Estimation of outer weights/loadings and path coefficients. These parameters are estimated through the simple regressions as depicted in the outer model equations.

Stage 3: Estimation of location parameters. In this stage, standardization is dropped, and values for the latent variables are estimated again as weighted sums of their indicators. Thus, PLS path modeling estimation always ends with expressing the latent variables as exact weighted sums of their associated indicators, in the metric of those observed variables.

The four steps in Stage 1 are repeated until the change in outer weights between two iterations drops below a predefined limit. The Stage 1 algorithm terminates after Step 1.1, delivering latent variable scores for all latent variables that are used to calculate loadings, outer weights, and inner regression coefficients in Stage 2 via single and multiple (partial) linear regressions. Location parameters are finally computed in Stage 3.

The evaluation of finally computed path modeling results is not straightforward since the nonparametric PLS approach does not provide a global goodness of fit criterion. As a consequence, Chin (1998) has put forward a catalog of criteria to assess PLS path modeling results, for example, by using standard errors that are obtained via bootstrapping. Henseler et al. (2009) describe in depth a systematic application of these criteria in a two-step process that involves (1) the assessment of the outer model and (2) the assessment of the inner model. In the first step, model assessment focuses on the measurement models. A systematic evaluation of PLS estimates reveals the measures' reliability and validity according to certain criteria that are associated with formative and reflective outer models, respectively. Only when this analysis provides evidence of sufficient reliability and validity is it necessary to evaluate the inner path model estimates in the second step. This kind of assessment also includes an evaluation of the predictive power of the model to reproduce the observed data. However, PLS' lack of a global optimization function and consequent lack of measures of global goodness of model fit definitely limit the use of PLS for theory testing (Hair, Ringle, & Sarstedt, 2011).

3. HETEROGENEITY IN PLS PATH MODELING VIA INTERACTION/NONLINEAR TERMS

After the estimation and evaluation of a PLS path model, complementary PLS analyses may focus on uncovering heterogeneity. As noted previously, one way to express heterogeneity is through the use of interaction or other nonlinear terms. Consider a simple regression model with two predictors, an interaction term and ratio scales (note that this does not hold for interval scales; Carte & Russell, 2003):

$$Y = \beta_0 + \beta_1 X + \beta_2 Z + \beta_3 XZ + \varepsilon \tag{6}$$

The connection to heterogeneity comes by way of the simple slopes implied by the model. The simple slope is found by taking the derivative of Y with respect to each predictor, separately, and thus represents the expected change in Y for a change in one predictor (Cohen et al., 2003). For a linear equation – an equation lacking nonlinear terms – the simple slope is a constant, equal to the regression coefficient. But with an interaction term in the model:

$$\frac{\partial Y}{\partial X} = \beta_1 + \beta_3 Z \tag{7}$$

and

$$\frac{\partial Y}{\partial Z} = \beta_2 + \beta_3 X \tag{8}$$

Thus, inclusion of the interaction term means that the simple slopes now depend on the values of the X and Z variables (PLS moderator analysis; Henseler & Fassott, 2010), so that these slopes can vary for each member of a population. Similarly, for a model with a quadratic term (PLS nonlinear analysis)

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \varepsilon \tag{9}$$

the simple slope becomes:

$$\frac{\partial Y}{\partial X} = \beta_1 + 2\beta_2 X \tag{10}$$

Thus, while each of these models represents one equation with one set of parameter values, the nonlinear terms imply simple slopes that vary across respondents, just as the values of the predictor variables vary across respondents.

Researchers have developed several approaches for the analysis of interaction effects between latent variables (e.g., in PLS moderator analysis; Henseler & Fassott, 2010; Chin, Marcolin, & Newsted, 2003). In situations where a moderator variable is categorical, observations are grouped into subsamples according to the moderator variable's modalities, leading to the multiple group analysis described below. The model may be estimated separately within each subsample, and the parameter estimates compared for significant differences. In the case of metric variables, the standard regression approach is to represent interaction terms by creating products of the main effect variables. The inclusion of the moderator variable's main effect is important to account for mean value changes in the dependent latent variable. Carte and Russell (2003) provide an in-depth discussion of common errors and their solutions in moderator analyses, which are also relevant for PLS path modeling. Moreover, the moderator analysis, complementing PLS path modeling results with additional findings on heterogeneity, may focus on a single indicator at a time for a more concise interpretation of outcomes.

Researchers have proposed a number of PLS-based approaches for modeling interaction and nonlinear terms (Fig. 1 provides an overview). Henseler and Chin (2010) compared approaches for modeling interactions in terms of point estimate accuracy, statistical power, and prediction accuracy. They concluded that the ''orthogonalizing approach'' (Little, Bovaird, & Widaman, 2006) is recommendable under almost all circumstances. This technique, adapted for SEM from regression (Lance, 1988), is designed to deal with the collinearity often found between main effect terms and interaction terms. In a regression interaction model, an indicator for the interaction of predictors X and Z is created by multiplying $X \cdot Z$. As part of this ''product indicator approach,'' the variables X and Z are usually centered (adjusted to have zero means) before multiplication, partly to aid interpretation but also to reduce correlation between main effects terms and interaction term (or between the linear term and the quadratic term, in a quadratic model). Even with this mean centering, however, if the distributions of the main effects are skewed, some correlation will remain (Cohen et al., 2003). The original extension of the product indicator approach to SEM with multiple indicators of each latent variable involved cross-multiplying each indicator of one main effect times each indicator of

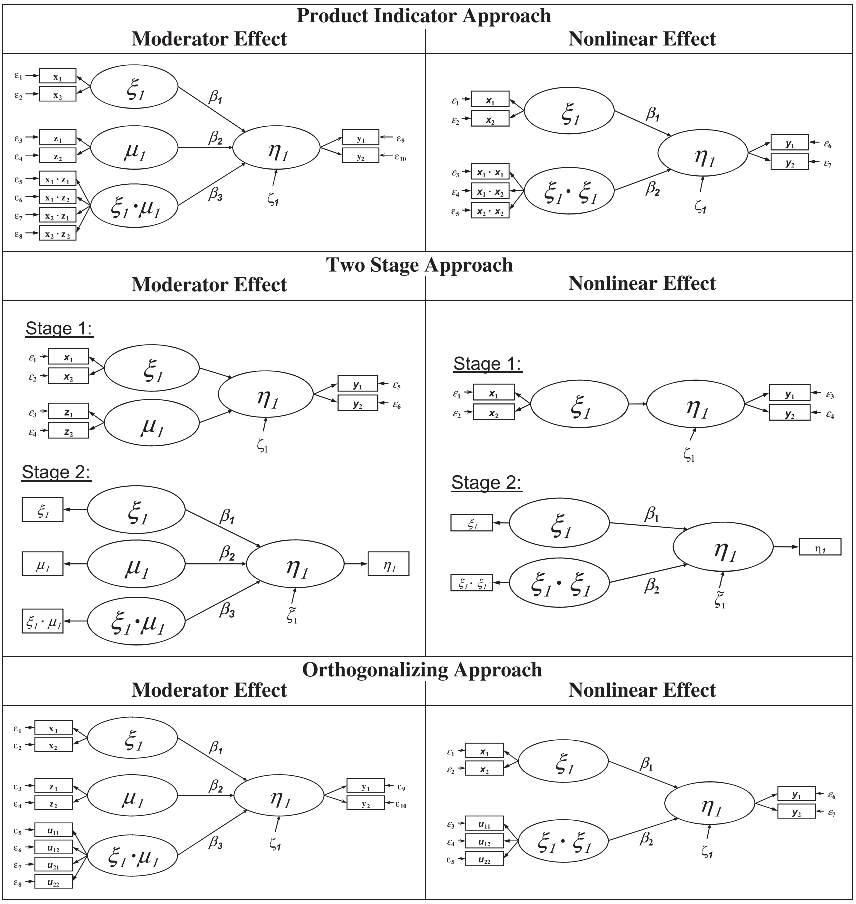

Fig. 1. Approaches for Modeling Interaction and Nonlinear Terms.

the other main effect, after first centering the indicators to reduce collinearity. As an alternative, the orthogonalizing approach cross-multiplies the indicators without centering. Then each of the resulting products is separately regressed on all of the original main effect indicators, retaining the residual. Once this is done for all of the resulting products, the residuals of these regressions are used as indicators of the interaction term, in analogy to the product indicator approach. Consequently, the interaction variable's indicators do not share any variance with the indicators of the "main effect" exogenous constructs. Fig. 1 illustrates the orthogonalizing approach, where

u_{11}, u_{12}, u_{21}, and u_{22} represent the residuals of the three regressions with the terms $x_1 \cdot z_1$, $x_1 \cdot z_2$, $x_2 \cdot z_1$, and $x_2 \cdot z_2$ as dependent variables. As pointed out by Henseler and Chin (2010), the latent interaction variable is orthogonal to the constituting latent variable because PLS calculates the latent variable scores as linear combinations of the respective indicators.

Additional research has looked for best methods for estimating PLS path models with other types of nonlinear terms (Fig. 1). Simulation studies on the use of the alternative approaches in PLS path modeling (Henseler, Wilson, & Dijkstra, 2007) show that the product indicator approach and the orthogonalizing approach described above should be used when parameter accuracy is a major issue of concern. Thus, these two approaches represent the best choice for hypothesis testing. However, when prediction represents the major or only purpose of an analysis, researchers should use a "two-stage approach." With this approach, researchers first estimate the model with only linear terms and capture the factor scores or case values for the latent variables. Then researchers create a single indicator for the nonlinear term by transforming the linear term factor score, and re-estimate the model including both the linear terms, with their indicators, and the nonlinear term with its single indicator.

Beyond these "purely PLS" approaches, innovations have introduced hybrid techniques that more easily accommodate nonlinear relations. Drawing on techniques familiar in data mining/automated data analysis, Hsu, Chen, and Hsieh (2006) propose an artificial neural network (ANN)-based SEM technique that can measure nonlinear relations by using different activity functions and layers of hidden nodes (Hackle & Westlund, 2000). The ANN-based approach is inspired by the way biological nervous systems process information and, thus, follows a completely different concept than the PLS approach. However, in essence, the approximation procedure is very similar to PLS, except that the ANN-based SEM technique can simultaneously measure inner and outer model relations. Results from simulation and empirical studies show that the ANN-based approach behaves very similarly to PLS path modeling.

Going still further, the NEUSREL package uses a Bayesian neural network (BNN) approach to search for interaction, quadratic and other higher order effects within models that specify only linear effects (Buckler & Hennig-Thurau, 2008). Thus, unlike standard PLS approaches discussed earlier, NEUSREL only requires the researcher to specify the linear part of the model. NEUSREL constructs starting values for its latent variables through principal component analysis, and then optimizes via the BNN approach, with the aim of maximizing variance explained while avoiding "overfitting" (weighting the equation with additional sample-specific predictive terms). Not surprisingly, the resulting model with nonlinear terms will often outperform a linear-only PLS model in terms of dependent variable variance explained. The estimation process is computationally intensive, unlike true PLS path modeling, which is normally speedy (Wold, 1982). NEUSREL is actually a set of software modules that exploit the capabilities of the MATLAB computational mathematics package, so users must currently have access to MATLAB in order to use NEUSREL. While such approaches as these last two do not constitute conventional PLS path modeling, the general trend of blurring the distinctions between analytical techniques will only accelerate as researchers focus on selecting the best tools to achieve their research mission.

4. MODELING HETEROGENEITY WITH A PRIORI GROUPS

Traditionally, heterogeneity in structural equation models is taken into account by assuming that observations can be assigned to segments a priori, on the basis of observable characteristics such as geographic or demographic traits (Wedel & Kamakura, 2000). In the case of a customer satisfaction analysis, this may be achieved by distinguishing high- and low-income user segments and carrying out multiple group comparisons.

Alternatively, sequential procedures have been proposed in which a researcher can partition the sample into segments by applying a clustering algorithm such as *k*-means on manifest or latent variable scores. However, different clustering algorithms yield different results, and, to date, there is little guidance on choosing the best procedure (Jedidi, Jagpal, & DeSarbo, 1997b). Furthermore, best practice would seem to call for clustering that is based on all available information, hence in relation to the defined model (Squillacciotti, 2010). Empirical studies and numerical experiments show that these "sequential" procedures – exploratory clustering followed by multiple group analysis – are not robust and perform poorly in terms of parameter recovery (Sarstedt & Ringle, 2010). Therefore, if researchers do not have an a priori rationale for distinguishing subgroups within a population, then latent class approaches, which allow for the identification and treatment of unobserved heterogeneity, seem to be a better choice. However, researchers may certainly be interested in differences between subgroups defined a priori, so there is certainly a place for a priori multiple group analysis in PLS path modeling. This method has long been available in most SEM packages to test hypotheses about differences across populations, in terms of both model structure (i.e., an entirely different model is postulated for each segment) and parameter values (i.e., the model remains the same across all segments, only the parameter values differ). In PLS path modeling, however, multiple group comparison is a rather new research field only experiencing ongoing development since the introduction of the first approach by Keil et al. (2000), who were interested in whether a certain population parameter β differed across two subpopulations ($\beta^{(1)} \neq \beta^{(2)}$) (also compare Chin, 2000).

In this test, the standard PLS algorithm is run first for each subsample, followed by bootstrapping to obtain standard errors of the parameter estimates. The test statistic depends on whether the standard errors of the parameter estimates differ significantly across the subsamples. If the estimates are equal, the test statistic is computed as follows:

$$t = \frac{b^{(1)} - b^{(2)}}{\sqrt{\frac{(n^{(1)}-1)^2}{(n^{(1)}+n^{(2)}-2)} \cdot \mathrm{se}(b^{(1)})^2 + \frac{(n^{(2)}-1)^2}{(n^{(1)}+n^{(2)}-2)} \cdot \mathrm{se}(b^{(2)})^2} \cdot \sqrt{\frac{1}{n^{(1)}} + \frac{1}{n^{(2)}}}} \sim t_{n^{(1)}+n^{(2)}-2} \quad (11)$$

Here, $b^{(1)}$ ($b^{(2)}$) denote the parameter estimates of the path coefficients in subsample one (two), $n^{(1)}$ ($n^{(2)}$) the number of observations in subsample one (two), and $\mathrm{se}(b^{(1)})$ ($\mathrm{se}(b^{(2)})$) the standard error of the path coefficient standard errors as resulting from the bootstrapping procedure. Sarstedt and Wilczynski (2009) describe the complementary approach for paired samples.

In cases where the standard errors are unequal, the test statistic takes the following form (Chin, 2000):

$$t = \frac{b^{(1)} - b^{(2)}}{\sqrt{\frac{n^{(1)}-1}{n^{(1)}}\mathrm{se}(b^{(1)})^2 + \frac{n^{(2)}-1}{n^{(2)}}\mathrm{se}(b^{(2)})^2}} \quad (12)$$

This test statistic is asymptotically t-distributed, and the degrees of freedom (df) are given as follows:

$$\mathrm{df} = \left\| \frac{\left(\frac{(n^{(1)}-1)}{n^{(1)}} \cdot \mathrm{se}(b^{(1)})^2 + \frac{(n^{(2)}-1)}{n^{(2)}} \cdot \mathrm{se}(b^{(2)})^2\right)}{\left(\frac{(n^{(1)}-1)\cdot \mathrm{se}(b^{(1)})^2}{n^{(1)2}} + \frac{(n^{(2)}-1)\cdot \mathrm{se}(b^{(2)})^2}{n^{(2)2}}\right)} - 2 \right\| \quad (13)$$

This approach requires that (1) the two models compared exhibit similar levels of goodness of fit, (2) the data are not too non-normal, and (3) measurement invariance holds (Chin, 2000).

It is obvious that this approach to multiple group comparisons with its inherent distributional assumptions does not fit the largely distribution-free character of the PLS path modeling approach. A more comprehensive approach for model comparison was introduced by Chin (2003) who applies a distribution-free data permutation test. This test seeks to scale the observed difference between a priori formed groups by comparing it to differences between groups that are randomly assembled from the data, without regard to a priori distinctions. In this permutation approach, the researcher first conducts an a priori multiple group analysis and computes the test statistic. Then the researcher generates an empirical distribution of test statistics. This process involves assigning cases randomly to two groups, estimating the model and calculating the test statistic for the parameter estimate difference. A sufficiently high number of iterations (e.g., 1,000) allows generating the test statistic distribution. The significance of the test statistic for the a priori groups is evaluated against this distribution of test statistics. If, for instance, more than 95% of the permutation test statistic exceeds the original statistic, the null hypothesis (the path coefficients are equal) should be rejected at $\alpha = .05$.

Likewise, Henseler (2007) proposed an alternative nonparametric procedure that was specifically designed for multiple group PLS analysis and that exhibits certain advantages in comparison to other approaches (Henseler et al., 2009). In this approach, the subsamples to be compared are exposed to separate bootstrap analyses, and the bootstrap outcomes serve as a basis for the hypothesis tests of group differences. Instead of relying on distributional assumptions, the new approach evaluates the observed distribution of the bootstrap outcomes. Given two subsamples with parameter estimate $b^{(1)}$ and $b^{(2)}$, the conditional probability $p(b^{(1)} > b^{(2)} | \beta^{(1)} \leq \beta^{(2)})$ has to be determined. Here, $\beta^{(1)}$ and $\beta^{(2)}$ represent the true population parameters of population one and two. A researcher would like to be sure that the probability of error is below a specified α-level before concluding that $\beta^{(1)}$ is larger than $\beta^{(2)}$. Using the parameter estimates of two subsamples from bootstrapping, the researcher can easily verify how probable a difference in parameters between two subpopulations is, and hence test their hypothesis with the following equation:

$$p\left(b^{(1)} > b^{(2)} \middle| \beta^{(1)} \leq \beta^{(2)}\right) = 1 - \sum_{g} \frac{\Theta\left((b_g^{(1)} + b^{(1)} - \bar{b}^{(1)}) - (b_g^{(2)} + b^{(2)} - \bar{b}^{(2)})\right)}{G^2} \tag{14}$$

In the above equation, G denotes the number of bootstrap samples, $b_g^{(1)}$ and $b_g^{(2)}$ the bootstrap parameter estimates per bootstrap sample, $\bar{b}^{(1)}$ and

$\bar{b}^{(2)}$ denote the means of the focal parameters over the bootstrap samples, and Θ stands for the unit step function, which has a value of one if its argument exceeds zero, otherwise zero. The superscript in parentheses marks the respective group. This equation states that G^2 (i.e., all possible) comparisons of bootstrap parameters have to be made.

This approach to PLS multiple group analysis does not require any distributional assumptions and is simple to apply by using the bootstrap outputs that are generated by the prevailing PLS implementations such as SmartPLS (Ringle et al., 2005b) and PLS-Graph (Soft Modeling, Inc., 1992–2002). Researchers can easily conduct the final calculations with available spreadsheet software applications.

As indicated above, one fundamental requirement for carrying out multiple group comparisons concerns the establishment of measurement invariance (Steenkamp & Baumgartner, 1998). According to Vandenberg & Lance (2000, p. 4), this step ''is a logical prerequisite to conducting substantive cross-group comparisons (e.g., tests of group mean differences, invariance of structural parameter estimates), but measurement invariance is rarely tested.'' Even though group comparisons require invariance of the elements of the measurement structure (i.e., factor loadings and measurement errors) and of response biases, researchers consciously or unconsciously assume that the structures of the measures that they compare across the groups are equal (Steinmetz, Schmidt, Tina-Booh, Wieczorek, & Schwartz, 2009). However, the validity of this assumption is critical for any conclusions about group-related differences. Little (1997) even states that one cannot claim that the construct is the same in the different groups unless the assumption of measurement invariance is confirmed. Tests for measurement invariance address four questions (Steinmetz et al., 2009, p. 600): ''Are the measurement parameters (factor loadings, measurement errors, etc.) the same across groups? Are there pronounced response biases in a particular group? Can one unambiguously interpret observed mean differences as latent mean differences? Is the same construct measured in all groups?''

Testing for measurement invariance has been broadly discussed in SEM literature – Vandenberg and Lance (2000) provide a review. Even though measurement invariance should be added to the well-established criteria of reliability, homogeneity, and validity when performing multiple group analysis, literature on PLS does not provide any suggestions to address this issue. An appropriate means of testing measurement model invariance in PLS path modeling addresses the four questions raised by Steinmetz et al. (2009) by using bootstrapping or permutation tests-based PLS multiple group analysis results (Keil et al., 2000; Chin & Dibbern, 2010). See Ringle, Sarstedt, and Zimmermann (2011) for an application.

Then again, an insistence on measurement invariance across groups carries its own assumption that the impact of group membership is limited to the structural parameters of the inner model. In many cases, this assumption is questionable or even implausible, and researchers should consider group membership effects on both structural and measurement parameters (Muthén, 2008). On the other hand, PLS path modeling is avowedly a method based on approximation, and a method designed for situations involving a less firmly established theoretical base (Wold, 1982). Therefore, it may be best for researchers simply to express appropriate caution in interpreting results from PLS path analysis involving multiple groups.

5. UNOBSERVED HETEROGENEITY IN PLS PATH MODELING

Modeling segments based on a priori information suffers from serious limitations. In many instances, substantive theory on the variables causing heterogeneity is unavailable or incomplete. Furthermore, observable characteristics such as gender, age, or usage frequency are often insufficient to capture heterogeneity adequately (Wedel & Kamakura, 2000). Heterogeneity is frequently unobservable and its true sources are unknown.

Different approaches, designed to capture and treat unobserved heterogeneity in PLS path models, have been proposed lately and reviewed by Sarstedt (2008a). These procedures generalize, for example, finite mixture modeling, typological regression, and genetic algorithm approaches to PLS path modeling. Fig. 2 shows a taxonomy of available procedures including key references.

Sánchez and Aluja (2006) introduced a decision tree–like structure approach in which segments are represented by the outer nodes of a segmentation tree. Their path modeling segmentation tree (PATHMOX) algorithm has been specifically designed to take into account external information, such as demographic variables, whose values are used to identify and differentiate segments, thus enhancing segment profiling. Using the external variables, PATHMOX makes two-way splits and estimates the PLS model for each subgroup thus defined. Just as a decision tree seeks to maximally discriminate, PATHMOX looks for the largest differences between subgroups in terms of the model parameter estimates (Trinchera, 2007). Once an optimal first split is

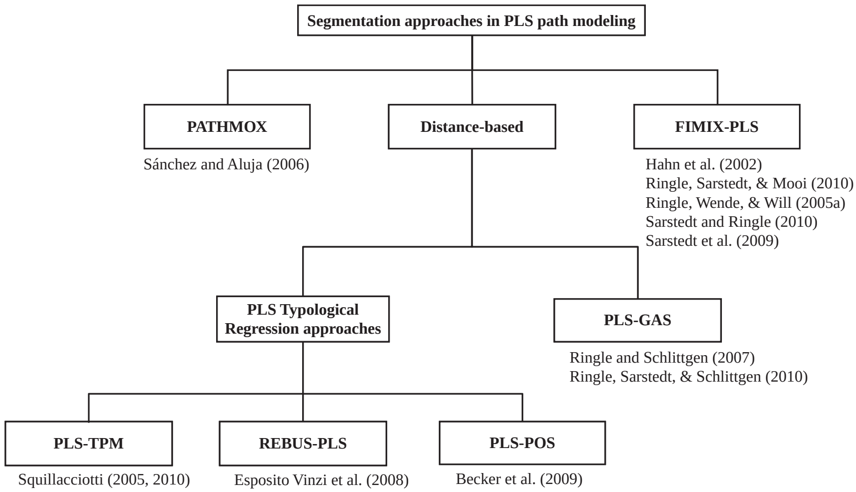

Fig. 2. Latent Class Approaches in PLS Path Modeling.

chosen, the algorithm then looks for further splits of those initial subgroups that again maximize differences in parameter estimates within subgroups. PATHMOX thus requires additional external data (Trinchera, 2007), and depends on the heterogeneity within the sample conforming to straightforward differences in the values of those external variables.

Another class of approaches uses distance measures to identify local models that are typical of specific segments. Squillacciotti (2005, 2010) has introduced the PLS typological path modeling (PLS-TPM) procedure, which has been designed for prediction-oriented path model segmentation to prevent imposing distributional assumptions on latent or manifest variables. This approach begins by estimating one global model for all observations and then clusters observations based on residuals relative to the global model. However, this measurement is not across the whole model, but relative to one single "target" block in the model (Trinchera, 2007). The researcher then chooses the number of classes or subgroups based on a dendrogram from this clustering of residuals. Next, individual models are estimated for each class. With each round of estimation, cases may be reassigned to different classes with the goal of minimizing an overall "distance" measure based on redundancy – the ability of the exogenous latent variables to account for the variance of the endogenous observed variable, as mediated by the endogenous latent variables (Trinchera, 2007) – across the entire data set, given the fixed number of classes. As noted above, the distance measure in PLS-TPM relates to a single target-dependent construct and associated observed variables. In models where the choice of a single target construct is unclear, PLS-TPM is less valuable (Trinchera, 2007).

Esposito Vinzi, Squillacciotti, Trinchera, and Tenenhaus (2008) have presented an improvement of PLS-TPM: response-based unit segmentation in PLS path modeling (REBUS-PLS), which has been designed to overcome methodological problems of the PLS-TPM approach by taking heterogeneity in endogenous and exogenous latent variables' inner and outer models into account. Here the distance measure (now labeled a "closeness measure" – see Trinchera, 2007) is a function of average communality (correlation of observed variables with their associated latent variables) and average structural R^2 (variance explained for the dependent latent variables) across the whole model. Otherwise, REBUS-PLS proceeds in much the same fashion as PLS-TPM. This alternative is included, for example, in the PLSPM package (Sánchez & Trinchera, 2010) for the R open source statistical programming language, and is available as a free download from the Comprehensive R Archive Network (CRAN; www.cran.r-project.org).

In this line of research, Ringle and Schlittgen (2007) consider the PLS segmentation task as an NP-complete data assignment problem of allocating observations to a set of segments. The solution time for NP-complete problems increases rapidly as the size of the problem grows, and there is no known efficient algorithm for speeding the process. Here, the size of the problem increases with the number of observations, the number of variables, and the number of classes. As it is impractical to test for all possible assignments, the authors propose a new kind of segmentation approach, PLS genetic algorithm segmentation (PLS-GAS), which uses a genetic algorithm to account for heterogeneity when estimating measurement and inner model relationships (Ringle, Sarstedt, & Schlittgen, 2010).

Finally, Becker, Ringle, and Völckner (2009) present the prediction-oriented segmentation method for PLS path modeling (PLS-POS). Like with PLS-GAS, the methodology has been designed to overcome several problems and limitations of existing PLS segmentation methods and shows very promising results in initial simulation studies.

An important improvement in the field of treating unobserved heterogeneity in PLS path models was presented by Hahn et al. (2002) and later advanced by Ringle, Wende, and Will (2005a), Ringle, Sarstedt, and Mooi (2010), and Ringle, Wende, and Will (2010). FIMIX-PLS combines the strengths of the PLS method with the advantages of the maximum likelihood estimation when deriving market segments with the help of finite mixture models. A finite mixture approach to model-based clustering assumes that the data originate from a source of several subpopulations (segments). Each segment is modeled separately and the overall population is a mixture of these segments (e.g., McLachlan & Peel, 2000; Frühwirth-Schnatter, 2006). That is, each data point x_i is taken to be a realization of the mixture density with K ($K < \infty$) segments, where

$$f_{i|k}(\mathbf{x}_i|\boldsymbol{\theta}_k) = \sum_{k=1}^{K} \rho_k f_{i|k}(\mathbf{x}_i) \tag{15}$$

with $\rho_k > 0$, $\forall k$, $\sum_{k=1}^{K} \rho_k = 1$, $f_{i|k}(\cdot)$ being a density function, and $\boldsymbol{\theta}_k$ depicting the segment-specific vector of unknown parameters for segment k. The set of mixing proportions ρ_k determines the relative mixing of the K segments in the mixture. Based on the fitted posterior probabilities of segment membership, a probabilistic clustering of the data into K clusters can be obtained. As a consequence, homogeneity is no longer defined in terms of a certain set of common scores but at a distributional level, whereas the magnitude of the relationships between latent variables may vary as a function of segment (Bauer & Curran, 2004). Consequently, finite mixture modeling enables researchers and practitioners to cope with heterogeneity in data by clustering observations and estimating parameters simultaneously, thus avoiding well-known biases that occur when models are estimated separately (Fraley & Raftery, 2002; Oh & Raftery, 2003).

6. THE FIMIX-PLS METHODOLOGY

This chapter uses the FIMIX-PLS method to illustrate the tasks and challenges involved in addressing latent homogeneity in PLS path modeling. This section takes an extended look at the FIMIX-PLS approach. The next section follows an empirical example – an application of FIMIX-PLS to the analysis of customer satisfaction and reputation data.

6.1. The FIMIX-PLS Algorithm

Building on the guiding articles by Jedidi et al. (1997a) and Hahn et al. (2002), a systematic application of FIMIX-PLS involves the steps illustrated in Fig. 3 (Ringle et al., 2010).

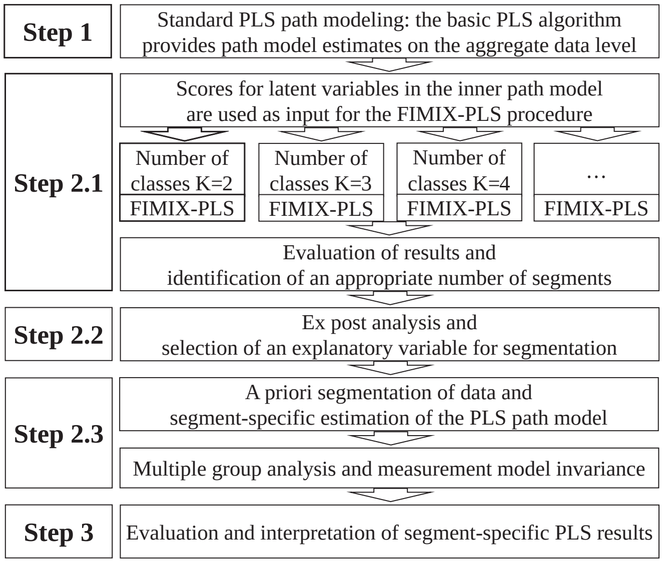

Fig. 3. Analytical Steps of FIMIX-PLS.

In Step 1 of the FIMIX-PLS methodology, a path model is estimated by using the standard PLS algorithm for path modeling with latent variables and the data of manifest variables in the outer models. The resulting scores of the latent variables in the inner path model are then employed to run the FIMIX-PLS algorithm in Step 2. The following equation expresses a modified presentation of the relationships:

$$B\eta_i + \Gamma\xi_i = \zeta_i \tag{16}$$

The segment-specific heterogeneity of path models is concentrated in the estimated relationships between latent variables. FIMIX-PLS captures this heterogeneity and calculates membership probabilities for each respondent belonging to each of the K segments (with the number K specified by the researcher). The segment-specific distributional function is defined as follows, assuming that η_i is distributed as a finite mixture of conditional multivariate normal densities $f_{i|k}(\cdot)$:

$$\eta_i \sim \sum_{k=1}^{K} \rho_k f_{i|k}(\eta_i|\xi_i, B_k, \Gamma_k, \Psi_k) \tag{17}$$

Substituting $f_{i|k}(\eta_i|\xi_i, B_k, \Gamma_k, \Psi_k)$ results in the following equation[1]:

$$\boldsymbol{\eta}_i = \sum_{k=1}^{K} \rho_k \left[\frac{1}{(2\pi)^{M/2}\sqrt{|\boldsymbol{\Psi}_k|}}\right] \exp\left\{-\frac{1}{2}((I-\mathbf{B}_k)\boldsymbol{\eta}_i + (-\boldsymbol{\Gamma}_k)\xi_i)'\boldsymbol{\Psi}_k^{-1}((I-\mathbf{B}_k)\boldsymbol{\eta}_i + (-\boldsymbol{\Gamma}_k)\xi_i)\right\} \tag{18}$$

The likelihood of the observed data is given by

$$L = \prod_{i=1}^{I}\left[\sum_{k=1}^{K} \rho_k \left[\frac{|\mathbf{B}_k|}{(2\pi)^{M/2}\sqrt{|\boldsymbol{\Psi}_k|}} \exp\left\{-\frac{1}{2}(\mathbf{B}_k\boldsymbol{\eta}_i + \boldsymbol{\Gamma}_k\boldsymbol{\xi}_i)'\boldsymbol{\Psi}_k^{-1}(\mathbf{B}_k\boldsymbol{\eta}_i + \boldsymbol{\Gamma}_k\boldsymbol{\xi}_i)\right\}\right]\right] \tag{19}$$

which can be expressed as

$$L = \prod_{i=1}^{I}\left(\sum_{k=1}^{K} \rho_k f(\eta_i|\xi_i, B_k, \Gamma_k, \Psi_k)\right) \tag{20}$$

Following from this, the log-likelihood of the model is

$$\ln\, L = \sum_{i=1}^{I} \ln\left(\sum_{k=1}^{K} \rho_k f(\eta_i|\xi_i, B_k, \Gamma_k, \Psi_k)\right) \tag{21}$$

Eq. (22) represents an expectancy-maximization or EM approach (Dempster, Laird, & Rubin, 1977), maximizing the log-likelihood ln L_c for the complete data problem:

$$\ln\, L_c = \sum_{i=1}^{I}\sum_{k=1}^{K} z_{ik}\; \ln(f(\eta_i|\xi_i, B_k, \Gamma_k, \Psi_k)) + \sum_{i=1}^{I}\sum_{k=1}^{K} z_{ik} \ln(\rho_k) \tag{22}$$

An EM formulation of the FIMIX-PLS algorithm (Fig. 4) is used in statistical computations to maximize the likelihood and to ensure convergence in this model. The expectation of Eq. (22) is calculated in the E-step, where z_{ik} is 1 if subject i belongs to segment k (0 otherwise). The segment size ρ_k, the parameters ξ_i, B_k, Γ_k, and Ψ_k of the conditional probability function (as results of the M-step), and provisional estimates (expected values), $E(z_{ik}) = P_{ik}$, for z_{ik} are computed according to Bayes' (1763/1958) theorem (E-step in Fig. 4).

Eq. (22) is maximized in the M-step (Fig. 4). Compared with the original finite mixture SEM technique (Jedidi et al., 1997a), this part of the FIMIX-PLS algorithm comprises the most important changes in order to fit the finite mixture approach to PLS. Initially, new mixing proportions ρ_k are calculated by the average of the adjusted expected values P_{ik} that result from the previous

```
// initial E-step
    set random starting values for P_ik; set last_lnLc = large number; set 0 < stop criterion < 1

// start with M-step

repeat do

begin

// the E-step starts here
    if Δ > stop criterion then
      begin
```

$$P_{ik} = \frac{\rho_k f_{i|k}(\eta_i \mid \xi_i, B_k, \Gamma_k, \Psi_k)}{\sum_{k=1}^{K} \rho_k f_{i|k}(\eta_i \mid \xi_i, B_k, \Gamma_k, \Psi_k)} \forall i, k$$

$$last_{lnL_c} = current_{lnL_c}$$

```
      end

// the M-step starts here
```

$$\rho_k = \frac{\sum_{i=1}^{I} P_{ik}}{I} \forall k$$

```
      determine B_k, Γ_k, Ψ_k, ∀k
      calculate current_lnLc
```

$$\Delta = \left| last_{lnL_c} \right| - \left| current_{lnL_c} \right|$$

```
end

until Δ < stop criterion
```

Fig. 4. The FIMIX-PLS Algorithm.

E-step. Thereafter, optimal parameters for B_k, Γ_k, and Ψ_k are determined by independent OLS regressions (one for each relationship between latent variables in the inner model). ML estimators of coefficients and variances are assumed to be identical to OLS predictions. The following equations are applied to obtain the regression parameters for latent endogenous variables:

$$Y_{mi} = \eta_{mi} \quad \text{and} \quad X_{mi} = (E_{mi}, N_{mi})' \tag{23}$$

$$E_{mi} = \begin{cases} \{\xi_1, \ldots, \xi_{A_m}\}, & A_m \geq 1, a_m = 1, \ldots, A_m \wedge \xi_{a_m} \text{ is the regressor of } m \\ & \varnothing \text{ else} \end{cases} \tag{24}$$

$$N_{mi} = \begin{cases} \{\eta_1, \ldots \eta_{B_m}\}, & B_m \geq 1, b_m = 1, \ldots, B_m \wedge \eta_{b_m} \text{ is the regressor of } m \\ & \varnothing \text{ else} \end{cases} \tag{25}$$

The closed-form OLS analytical formula for τ_{mk} and ω_{mk} is expressed as follows:

$$\tau_{mk} = ((\gamma_{a_m mk}), (\beta_{b_m mk})) = [X'_m P_k X_m]^{-1}[X'_m P_k Y_m] \tag{26}$$

$$\omega_{mk} = \text{cell } (m \times m) \text{ of } \Psi_k = \frac{(Y_m - X_m \tau_{mk})'((Y_m - X_m \tau_{mk})P_k)}{I \rho_k} \tag{27}$$

As a result, the M-step determines new mixing proportions ρ_k, and the independent OLS regressions are used in the next E-step iteration to improve the outcomes of P_{ik}. The EM algorithm stops whenever ln L_c no longer improves, and an a priori specified convergence criterion is reached.

A substantial body of simulation evidence shows that FIMIX-PLS reliably identifies a priori formed segments in idealized situations with almost no noise in the distinctive, artificially generated sets of data (Ringle et al., 2005a). FIMIX-PLS also appropriately classifies two a priori generated groups of artificial data according to their distinctive inner PLS path model coefficients when the noise and, thus, the fuzziness of segments increases considerably (Ringle et al., 2010). Esposito Vinzi, Ringle, Squillacciotti, and Trinchera (2007) reveal that FIMIX-PLS also performs extremely well in situations with unbalanced segments and non-normal data. Likewise, Tenenhaus, Mauger, and Guinot (2010) show that the endogenous latent variables approach a normal distribution, even if both the manifest and exogenous latent variable scores are far from normal (Hair et al., 2011). Consequently, the potential limitation of FIMIX-PLS that the approach imposes a distributional assumption on the endogenous latent variables has no far-reaching practical implications.

6.2. Identifying an Appropriate Number of Segments

When applying finite mixture models to empirical data, the actual number of segments K is usually unknown. Retaining the true number of segments is, however, crucial, as many managerial decisions rely on this decision (Sarstedt, 2008b). Various tests and heuristics have been proposed to determine the number of segments, but this so-called model selection problem is a longstanding and still unresolved issue with the least satisfactory statistical treatment.

An obvious way to determine the number of segments in the mixture model is to test the null hypothesis of K segments against the alternative hypothesis of $K+1$ segments by carrying out a standard likelihood ratio test (LRT). However, as pointed out by several authors (e.g., McLachlan & Peel, 2000; Wedel & Kamakura, 2000), the LRT is unsuitable in a finite mixture framework. Although the two models are conceptually nested, the model with only K segments constrains membership probabilities for the $K+1$ segment to 0, a boundary point (Muthén, 2008). As a result, the test statistic does not follow a central χ^2-distribution under the null hypothesis.

In light of this problem, researchers frequently revert to a heuristic approach in the form of model selection criteria to determine the number of segments that can be categorized into information and classification criteria (McLachlan & Peel, 2000; Sarstedt et al., 2011). Information criteria are based on a penalized form of the likelihood, as they simultaneously take the goodness of fit (likelihood) of a model and the number of parameters used to achieve that fit into account. These criteria therefore correspond to a penalized likelihood function (i.e., the negative log-likelihood plus a penalty term that increases with the number of parameters and/or the number of observations; Sarstedt, 2008b; Sarstedt et al., 2011). Information criteria generally favor models with large log-likelihood values and few parameters, and are scaled so that a lower value represents a better fit.

In keeping with substantive theory, applied researchers usually use a combination of criteria to guide their decision. Popular criteria include the Akaike information criterion (AIC; Akaike, 1973), the modified AIC with factor 3 (AIC_3; Bozdogan, 1994), the consistent AIC (CAIC; Bozdogan, 1987), and the Bayesian information criterion (BIC; Schwarz, 1978). Although these heuristics account for overparameterization through the integration of a penalty term, they do not ensure the sufficient separation of the segments in the selected solution. As the targeting of markets requires the segments to be differentiable – the segments should be conceptually distinguishable and should respond differently to different marketing mix elements – this point is of great practical interest.

Within this context, the normed entropy statistic (EN; Ramaswamy, DeSarbo, Reibstein, & Robinson, 1993) is a critical criterion for analyzing whether segment-specific FIMIX-PLS results produce well-separated clusters:

$$\mathrm{EN} = 1 - \frac{\sum_{i=1}^{I} \sum_{k=1}^{K} -P_{ik} \ln(P_{ik})}{I \ \ln(K)} \tag{28}$$

The EN is limited to between 0 and 1, and the quality of the separation of derived classes is commensurate with the increase in EN. Applications of FIMIX-PLS furnish evidence that EN values of below .5 indicate fuzzy class memberships that prevent meaningful segmentation and limit the applied value of the segmentation exercise.

Sarstedt et al. (2011) evaluate the performance of several model selection criteria including the ones described above. Their results show that a joint consideration of AIC_3 and CAIC proves valuable in most settings as these criteria indicate the correct number of segments in 85% of all cases where both point to the same segment number. Likewise, AIC's pronounced tendency to overestimate the correct number of segments as well as ICL-BIC's (Biernacki, Celeux, & Govaert, 2000) tendency to underestimate the correct number of segments can guide model selection in practical applications. Conversely, entropy-based measures proved unsuitable for deciding on the number of segments. However, to ensure managerially relevant segmentation outcomes, researchers should not rely solely on AIC_3 and CAIC (or AIC and ICL-BIC in the context described above). Researchers should also ensure that segments are sufficiently separated.

Segment size – the number of cases falling into each segment – is another useful indicator to prevent the selection of too many classes. The EM algorithm always converges to the prespecified number of K segments. Compared with alternative procedures (e.g., the Newton–Raphson method), this is advantageous, but when an analysis selects an "extra" segment, some observations are "forced" to belong to the extraneous segment even though they truly fit in another segment. In such situations, the additional segment is usually very small and accounts for only a marginal portion of heterogeneity in the overall data set. Such tiny subsets of the sample are also unlikely to translate into meaningful market segmentation opportunities.

6.3. Ex Post Analysis

In situations where FIMIX-PLS results indicate that in the overall data set, heterogeneity can be reduced by fitting a finite mixture model with K segments, turning this statistical finding into actionable recommendations requires that the researcher interprets the subgroups in terms of managerially meaningful variables. This involves an ex post search for explanatory variables as noted in Step 2.2 (ex post analysis) in Fig. 3. These variables must imply a similar grouping of respondents to that indicated by the FIMIX-PLS results.

Moreover, the ex post analysis is important from a methodological point of view. Measurement modes can only be updated through the ex post analysis, thus leading to segment-specific outer model parameter estimates.

Correspondingly, data are segmented and used as new input for segment-specific path model computations with PLS. This process generates differentiated PLS modeling results in both inner and outer models and facilitates multiple group PLS analyses (Chin & Dibbern, 2010). This kind of analysis is essential for exploiting FIMIX-PLS findings for PLS path modeling and is the most challenging analytical step to accomplish.

Ramaswamy et al. (1993) suggest a systematic search to uncover explanatory variables that fit well with the results in terms of segment affiliations. Specifically, the authors suggest estimating the following model for each segment k:

$$Q_{ik} = \sum_{c}^{C} Z_{ic}\omega_{ck} + v_{ik} \qquad (29)$$

where $Q_{ik} = (\ln(P_{ik}/P_{i\cdot})$, and $P_{i\cdot} = (\prod_{k=1}^{K} P_{ik})^{1/K}$ is the geometric mean of the probabilities of segment membership, Z_{ic} the value of the explanatory variable c for observation i, ω_{ck} the impact coefficient for variable c for segment k, and v_{ik} a random normal disturbance (Ramaswamy et al., 1993). Likewise, researchers may apply a discriminant or logistic regression analysis to determine which predictor variables contribute most of the intersegment differences. Ringle, Sarstedt, and Mooi (2010) and Sarstedt, Ringle, and Schwaiger (2009) apply a CHAID procedure to identify potential explanatory variables, but other tree algorithms for classification problems such as QUEST may likewise be used. Different from Ramaswamy et al.'s (1993) approach that relies on (modified) probabilities of segment membership, Ringle, Sarstedt, and Mooi (2010) use the segment affiliation variable z_{ik} as the dependent variable in these types of analysis. For the most part, a logical search focuses on managerial interpretation of results. In this case, relevant variables for explaining the expected differences in segment-specific PLS path model computations are examined for their ability to form groups of observations that match FIMIX-PLS results. Finally, PLS multiple group analyses allow comparing segment-specific path model estimates and testing whether the path coefficients differ significantly between the segments.

7. EMPIRICAL EXAMPLE

An exemplary model, which analyzes the effects of corporate-level marketing activities on corporate reputation and, finally, on customer satisfaction and loyalty, is used to demonstrate the applicability of FIMIX-PLS. In keeping with the analysis by Eberl (2010), it can be argued that marketing activities do not influence customers' loyalty decisions directly but do so via the mediating construct of corporate reputation, which is itself conceptualized as having two dimensions (Schwaiger, 2004). One dimension comprises all of the company's cognitive evaluations (*competence*), whereas the second dimension captures affective judgments (*likeability*). *Competence* and *likeability* were each operationalized by means of three indicators identified as exchangeable and thus treated reflectively.

Past research identified four exogenous driver constructs – *quality*, *performance*, *attractiveness*, and *corporate social responsibility (CSR)* – which have been shown to be robust across different data sets, countries, and industries (e.g., Schwaiger, 2004; Eberl & Schwaiger, 2005; Wilczynski, Sarstedt, & Melewar, 2009). These constructs were thus operationalized using a total of 21 formative indicators (Gudergan, Ringle, Wende, & Will, 2008) that measure the levers of corporate-level marketing activities. In keeping with the analysis by Eberl (2010), it was hypothesized that the two dimensions of corporate reputation relate to *customer satisfaction*, while *likeability* also influences *customer loyalty* directly. *Customer satisfaction* and *loyalty* were thus operationalized by means of reflective measures that are well known from empirical marketing studies (Zeithaml, Berry, & Parasuraman, 1996). Fig. 5 illustrates the path model under consideration.

The PLS path model analysis draws on four major service providers in Germany's mobile communications market, including two large providers who had a combined market share of more than 74% in 2005. Data were collected by means of computer-assisted telephone interviews from a representative sample of the German overall population. Each respondent rated the reputation and driver construct indicators on seven-point Likert scales. Satisfaction and loyalty were surveyed in respect of the interviewees' own service providers. The survey also tested several sociodemographic characteristics. The sample data set presented in this chapter comprises $N = 180$ subjects.[2]

As depicted in Fig. 3, the basic PLS algorithm (Wold, 1982; Lohmöller, 1989) is applied to estimate the overall model by means of SmartPLS 2.0 (Ringle et al., 2005b) in Step 1. We follow the suggestions for a PLS path model evaluation by Chin (1998) and apply the systematic approach for this

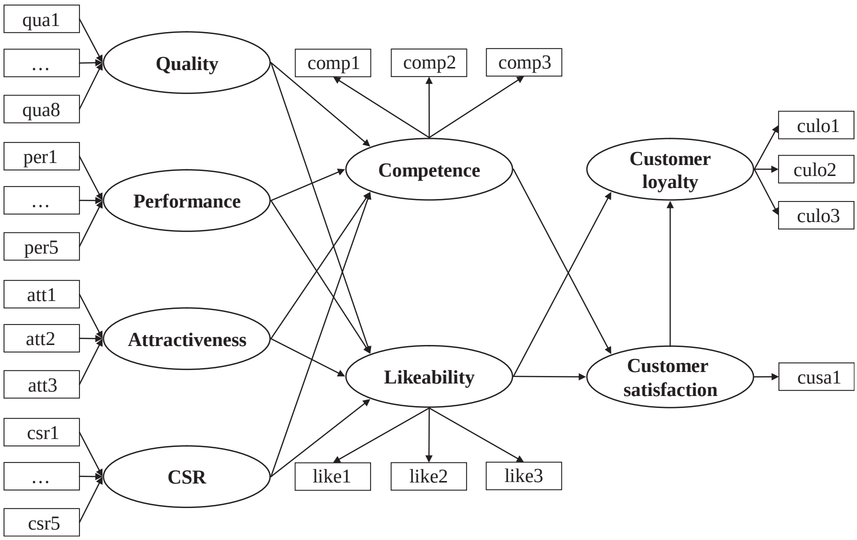

Fig. 5. Research Model for the Empirical Example.

kind of analysis recommended by Henseler et al. (2009). All the minimum requirements for the outer models and inner model are met. For example, the loadings of the reflective set of outer relationships clearly range above .7; the values of both the composite reliability ρ_c and Cronbach's α are uniformly high around .8, thus meeting the stipulated thresholds (Nunnally & Bernstein, 1994). Likewise, all reflectively measured constructs exhibit discriminant validity (Fornell & Larcker, 1981). With regard to this global model, the R^2 values of the constructs *competence*, *likeability*, and *customer loyalty* are very acceptable, indicating that both dimensions of corporate reputation are good predictors of loyalty (Table 1). As indicated by Eberl (2010), the rather low value of *customer satisfaction* is not surprising, as intangible assets such as corporate reputation represent only one of the numerous determinants of this construct.

An evaluation of the path coefficients' significance draws on t-values that were calculated via a bootstrapping procedure (5,000 subsamples; individual-level sign changes; Henseler et al., 2009). The analysis of the interrelations between the constructs reveals that most path relationships are significant at a level of .05 (see Table 1). The analysis shows that corporate reputation's affective dimension clearly dominates in explaining customer satisfaction. This implies that in the German telecommunication

Table 1. Analytical Results of the Aggregate-Level Data Analysis.

	PLS Path Coefficients
Quality → competence	.383** (7.502†)
Performance → competence	.375** (7.680†)
Attractiveness → competence	.003 (.218†)
CSR → competence	.113** (3.031)
Quality → likeability	.397** (7.843†)
Performance → likeability	.085* (1.836†)
Attractiveness → likeability	.184** (3.718†)
CSR → likeability	.202** (4.290)
Competence → customer satisfaction	.155** (2.721†)
Likeability → customer satisfaction	.454** (8.761†)
Customer satisfaction → customer loyalty	.522** (14.960†)
Likeability → customer loyalty	.348** (9.800†)
ρ_k	1.0
R^2 (competence)	.641
R^2 (likeability)	.585
R^2 (customer satisfaction)	.316
R^2 (customer loyalty)	.593

(*) Significant at $p < .10$; (**) significant at $p < .05$; (†) bootstrap t-value.

market, investments in marketing activities to enhance the perception of corporate competence do not pay off in terms of an increase in customer satisfaction, and thus loyalty. This finding is not surprising given the intangible nature of this industry's products, which makes these products hard to distinguish (Eberl, 2010). Much of the strategic benefit comes from a differentiation with regard to the affective dimension of corporate reputation.

However, due to unobserved heterogeneity in the data, this aggregate-level analysis might be misleading and the usage of FIMIX-PLS could

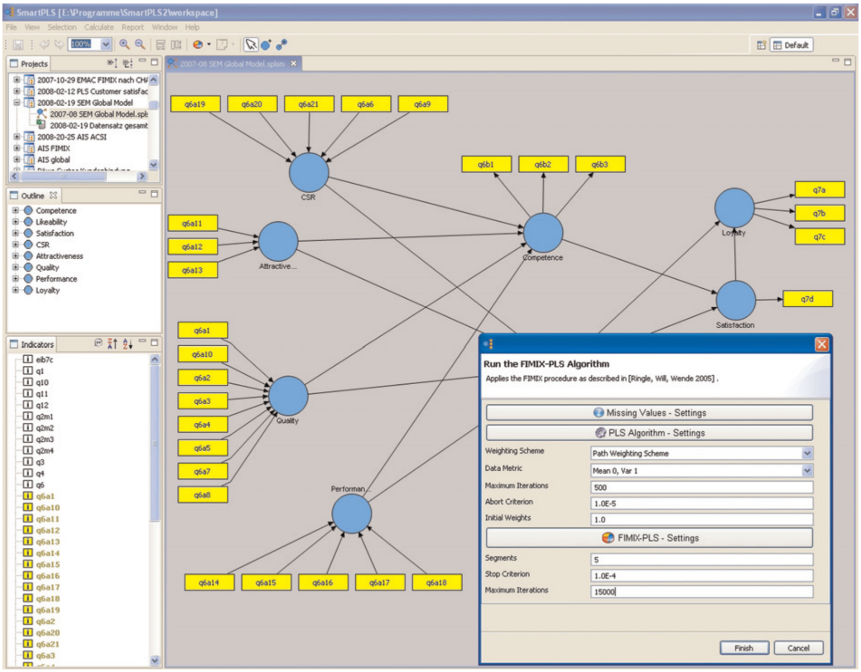

Fig. 6. Screenshot of SmartPLS with FIMIX-PLS Settings.

provide further differentiated results. The FIMIX-PLS module SmartPLS 2.0 (Ringle, Wende, & Will, 2005b) was applied to segment observations based on the estimated latent variable scores (Step 2.1 in Fig. 3). Fig. 6 shows a screenshot of the module in SmartPLS.

Initially, FIMIX-PLS results were computed for two segments. Thereafter, the number of segments was increased successively. A comparison of the segment-specific information and classification criteria reveals that in almost all cases, the minimum criterion value for a two-segment solution was achieved (Table 2). Most notably, AIC_3 and CAIC both indicate the same number of segments, which provides strong support for a two-segment solution (Sarstedt et al., 2011). Across all solutions, EN achieved values clearly above .6, thus indicating well-separated segments. The choice of two segments is additionally supported by the FIMIX-PLS segment sizes as well as the a priori classification on the basis of probabilities of membership (Table 3). For example, in the five-segment solution, one segment comprises only 7.2% of all observations.

Table 2. Model Selection.

	$K=2$	$K=3$	$K=4$	$K=5$	$K=6$
AIC	1,557.822	1,580.439	1,631.934	1,590.232	1,616.995
BIC	1,663.189	1,740.087	1,845.862	1,858.440	1,939.483
CAIC	1,696.189	1,790.087	1,912.862	1,942.440	2,040.483
AIC_3	1,590.822	1,630.439	1,698.934	1,674.232	1,717.995
EN	.610	.658	.649	.748	.731

Table 3. Relative Segment Sizes for Different Numbers of Segments.

	$k=1$ (%)	$k=2$ (%)	$k=3$ (%)	$k=4$ (%)	$k=5$ (%)	$k=6$ (%)	Sum (%)
$K=2$	38.9	61.1					100
$K=3$	40.5	43.9	15.6				100
$K=4$	45.0	11.6	16.7	26.7			100
$K=5$	33.9	23.9	12.8	7.2	22.2		100
$K=6$	24.4	23.9	36.1	5.0	5.0	5.6	100

To estimate the segment-specific path coefficients in the outer models, observations were assigned to the two segments according to each observation's maximum a posterior probability of segment membership. Table 4 provides an overview of the FIMIX-PLS results in respect of two segments, the mixing proportions of each segment ρ_k, and the R^2 value of each endogenous latent constructs. The empirical example also draws on reflective latent variables that require measurement model invariance to be tested before multiple group analyses can be carried out. To test for measurement invariance, we use the nonparametric, permutation test–based multiple group analysis by Chin and Dibbern (2010) to test – among other criteria (Steenkamp & Baumgartner, 1998) – if the group-specific (a) outer loadings and (b) the AVE/ρ_c outcome per reflectively measured latent construct differ significantly (see also Ringle et al., 2011). According to this analysis, measurement model invariance is not an issue in the PLS multiple group analysis that builds on the FIMIX-PLS results. As a consequence, we carried out a multiple group comparison using Henseler's (2007) nonparametric approach in the next step of the analysis.

When comparing the aggregate-level analysis with the results derived from the FIMIX-PLS procedure, one finds that all endogenous constructs exhibit increased R^2 values, ranging between 1.2% (likeability) and 39.7% (customer satisfaction) higher than that in the global model. Likewise, the

Table 4. Analytical Results of the FIMIX-PLS Analysis.

Data Analysis Strategy	FIMIX-PLS			Ex Post Analysis		
	$k=1$	$k=2$	|Diff|	$k=1$	$k=2$	|Diff|
Quality → competence	.682** (27.754†)	.232** (4.160†)	.450**	.701** (21.085†)	.167** (3.155†)	.534**
Performance → competence	.521** (17.773†)	.297** (5.229†)	.224**	.281** (6.511†)	.471** (9.369†)	.190*
Attractiveness → competence	−.041* (2.374†)	.060 (1.499†)	.101**	−.133** (3.937†)	.078** (2.125†)	.211**
CSR → competence	.231** (11.332†)	.247** (5.922†)	.016**	.040 (1.323†)	.161** (4.248†)	.121
Quality → likeability	.516** (11.458†)	.343** (7.184†)	.173*	.487** (11.447†)	.383** (6.490†)	.104
Performance → likeability	.107** (2.674†)	.092** (2.042†)	.015	.029 (1.056†)	.072 (1.455†)	.043
Attractiveness → likeability	.246** (7.361†)	.178** (3.633†)	.068	.359** (8.355†)	.062 (1.426†)	.297**
CSR → likeability	.142** (3.390†)	.193** (4.329†)	.051	.074* (1.909†)	.291** (6.920†)	.217**
Competence → customer satisfaction	.834** (43.310†)	−.134** (2.538†)	.968**	.192** (3.124†)	.131** (2.552†)	.061
Likeability → customer satisfaction	.114** (5.281†)	.461** (10.330†)	.347**	.325** (5.978†)	.526** (10.800†)	.201**
Customer satisfaction → customer loyalty	.748** (25.086†)	.480** (16.335†)	.268**	.423** (9.827†)	.548** (17.069†)	.125*
Likeability → customer loyalty	.187** (6.142†)	.370** (12.998†)	.183**	.483** (13.654†)	.309** (9.431†)	.174**
ρ_k	.389	.611		.361	.639**	
R^2 (competence)	.649			.666		
R^2 (likeability)	.601			.597		
R^2 (customer satisfaction)	.440			.319		
R^2 (customer loyalty)	.627			.599		

(*) Significant at $p<.10$; (**) significant at $p<.05$; (†) bootstrap t-values.
Significance testing of |diff| uses Henseler's (2007) approach for multiple group analyses.

relative importance of the antecedents of corporate reputation differs quite substantially between the two segments. More importantly, there are different drivers of customer satisfaction in each segment. Although in the second segment, satisfaction is primarily driven by likeability (analogous to the global model), the cognitive dimension clearly dominates in explaining satisfaction in the first segment with an extremely high path coefficient of .834. With a single exception, all paths are significant at a level of .05. The multiple group comparison shows that most path coefficients differ significantly across the two segments. These results demonstrate that an aggregate-level analysis can in fact be misleading and that by grouping segments of customers, important implications for marketing mix programs can be achieved.

The next step involves the identification of explanatory variables that best characterize the two uncovered segments (Step 2.2 in Fig. 3). We consequently applied the QUEST algorithm (Loh & Shih, 1997) using IBM SPSS Answer Tree 3.1 on the covariates to assess if splitting the sample based on sociodemographic variables' modalities leads to a significant discrimination between the dependent measures of segment affiliation.[3] The analysis showed that only the variable that captured the interviewee's own service provider showed any potential for a meaningful segmentation. This result is also supported by binary logistic regression results showing that this variable is the only one that has a significant effect on the variable (Wald test, 5.729, $p = .017$). In accordance with these outcomes, segment one comprises customers of one of the two large mobile service providers, whereas the second segment consists of the remaining providers. This result is surprising, as one might expect differences between the customers of the two larger and smaller service providers. The resulting segment sizes are almost equal to the previous analysis, and cross-tabulation shows that the classification corresponds to 56.1% with the FIMIX-PLS classification. Evaluation of the PLS estimates (Chin, 1998; Henseler et al., 2009) of these two a priori segmented sets of data confirms the satisfactory results. All R^2 values range higher than those in the global model, indicating an improved goodness of fit (Table 4).

Segment-specific path analyses show similar results when compared with the FIMIX-PLS with regard to the driver constructs. For example, the influence of quality on competence differs substantially between the two segments ($k = 1$: .701; $k = 2$: .167), whereas the relationship is balanced in the global model (.383). Most differences between the segments in the ex post analysis are significant at .05, indicating that the segment-specific influence of corporate reputation's antecedents varies in the population.

However, segment-specific influences on the key construct customer satisfaction are less pronounced compared with the FIMIX-PLS analysis. Similar results are obtained regarding the antecedents of corporate reputation, where several differences in path coefficients are significant. This suggests that customers react differently to corporate-level marketing activities. For example, changes in the driver constructs should thus be targeted individually. From a marketing viewpoint, this result is important for companies to successfully enhance customer satisfaction via the mediating affective dimension corporate reputation.

8. SUMMARY AND CONCLUSION

Unobserved heterogeneity and measurement errors are epidemic problems in social sciences. Jedidi et al. (1997a) have addressed these problems with respect to SEM. The aim of this chapter has been to review options for dealing with heterogeneity in PLS path modeling. Currently, a variety of approaches and techniques are available, with a similar variety of drawbacks and limitations. This chapter examined the FIMIX-PLS approach in some detail (Hahn et al., 2002), but the key conclusion to be drawn from this manuscript is that researchers need to address heterogeneity in their inner models. For researchers and marketers, heterogeneity remains both a challenge and an opportunity. If ignored, heterogeneity can produce misleading results, giving rise to a model with parameter estimates that may be incorrect for every single respondent. If acknowledged, heterogeneity may well force researchers to require larger sample sizes so that models can be reliably estimated for all meaningful segments. At the same time, dealing with heterogeneity offers researchers the opportunity to better fit inner models to the patterns of variance actually observed in the data, and perhaps to understand respondents at a finer level. For marketers, of course, segmentation – along with targeting and positioning – is a core function and capability, and a key to profitability in a challenging competitive environment. Global competition and pressures toward commoditization continually threaten to reduce any market to the least common denominator. Understanding heterogeneity in the marketplace puts marketers in a position to best understand the needs not only of the niche customer, but also of every customer. Advances in PLS path modeling tools for dealing with heterogeneity make it feasible for researchers to routinely take this issue into account in their research. We can only expect that this issue will gain even greater prominence in the years ahead.

NOTES

1. Note that this description of the FIMIX-PLS procedure differs from Hahn et al.'s (2002) original presentation that exhibits several inconsistencies (see Sarstedt, Becker, Ringle, & Schwaiger, 2011).
2. The data of this analysis are available from the authors on request.
3. Likewise, Shih's (2005) software program QUEST may be used, which is freely available at http://www.stat.wisc.edu/~loh/quest.html.

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APPENDIX

Table A.1. Explanation of Symbols.

A_m	Number of exogenous variables as independent variables in regression m
a_m	Exogenous variable a_m with $a_m = 1, \ldots, A_m$
B_m	Number of endogenous variables as independent variables in regression m
b_m	Endogenous variable b_m with $b_m = 1, \ldots, B_m$
$\gamma_{a_m mk}$	Regression coefficient of a_m in regression m for segment k
$\beta_{b_m mk}$	Regression coefficient of b_m in regression m for segment k
τ_{mk}	$((\gamma_{a_m mk}), (\beta_{b_m mk}))'$ vector of the regression coefficients
ω_{mk}	Cell $(m \times m)$ of ψ_k
c	Constant factor
$f_{i\|k}(\cdot)$	Probability for case i given a segment k and parameters $(\cdot)$
I	Number of cases or observations
i	Case or observation i with $i = 1, \ldots, I$
J	Number of exogenous variables
j	Exogenous variable j with $j = 1, \ldots, J$
K	Number of classes
k	Segment or class k with $k = 1, \ldots, K$
M	Number of endogenous variables
m	Endogenous variable m with $m = 1, \ldots, M$
N_k	Number of free parameters defined as $(K-1) + KR + KM$
P_{ik}	Probability of membership of case i to segment k
R	Number of predictor variables of all regressions in the inner model
S	Stop or convergence criterion
V	Large negative number
X_m	Case values of the independent variables for regression m
Y_m	Case values of the independent variables for regression m
z_{ik}	$z_{ik} = 1$, if the case i belongs to segment k; $z_{ik} = 0$ otherwise
ζ_i	Random vector of residuals in the inner model for case i
η_i	Vector of endogenous variables in the inner model for case i
ξ_i	Vector of exogenous variables in the inner model for case i
$BM \times M$	Path coefficient matrix of the inner model
$\Gamma M \times J$	Path coefficient matrix of the inner model
Δ	Difference of current$_{\ln L}$ and last$_{\ln L}$
$B_k\ M \times M$	Path coefficient matrix of the inner model for latent segment k
$\Gamma_k\ M \times J$	Path coefficient matrix of the inner model for latent segment k
$\Psi_k\ M \times M$	Matrix for latent segment k containing the regression variances
ρ	$(\rho_1, \ldots, \rho_K)$, vector of the K mixing proportions of the finite mixture
ρ_k	Mixing proportion of latent segment k
Multiple group comparison	
$\beta^{(1)}/\beta^{(2)}$	Vector of true population parameters of population one/two
$b^{(1)}/b^{(2)}$	Vector of parameter estimates of path coefficients in subsample one/two
$n^{(1)}/n^{(2)}$	Number of observations in subsample one/two
g	Bootstrap sample with $g = 1, \ldots, G$

Table A.1. (*Continued*)

G	Number of bootstrap samples
Θ	Unit step function; $\Theta = 1$ if argument exceeds zero; $\Theta = 0$ otherwise
Ex post analysis (Ramaswamy et al., 1993)	
ω_{rk}	Impact coefficient for explanatory variable r for segment k
C	Number of explanatory variables
C	Explanatory variable c with $c = 1, \ldots, C$
$P_{i\cdot}$	Geometric mean of the probabilities of segment membership
Q_{ik}	Dependent variable in the ex post analysis
Z_{ir}	Case values of the dependent variables for regression r of individual i
v_{ik}	Random normal disturbance term